MORGAN STATE FOOD RESOURCE CENTER

The pantry was full. The line was empty handed.

So I built the system that would have served me.

The line I stood in

Students at Morgan State waited 40 minutes at the Food Resource Center, signed a paper sheet, and walked away when the manager could not match supply to demand in real time. I had been in that line. The week I started building, 312 students were turned away from food that was literally on the shelf behind them.

A pantry that talks to itself

1

Real time inventory and student request portal

Students request what they need. The pantry shows what is on the shelf right now.

2

Manager dashboard with auto matching

Requests pair with stock by category, quantity, and dietary tag without paper.

3

Wait time analytics for grant reporting

Administrators export the data USDA and the university already ask for.

Built in two weeks

FRONTEND

React, Vite, TypeScript

BACKEND

Express, Node 20, Drizzle ORM

DATABASE

PostgreSQL on Neon

AUTH

Replit Auth, role based

REAL TIME

WebSocket inventory sync

HOSTING

Replit, custom domain

The pilot week

40 min

Average wait, before

6 min

Average wait, after

312

Students served, week one

92%

Inventory match rate

Three lessons

- Build for the line you stood in. Your memory is the user research.
- Real time is the product. Without WebSockets the dashboard lies.
- Free tools (Neon, Replit, Drizzle) ship before enterprise stacks finish scoping.

Where it goes

Rolling out to all four HBCU food pantries in the DMV by fall 2026 with shared analytics and a USDA aligned reporting layer for grant applications. Open sourcing the entire stack so any campus can fork it.

The bottleneck was never supply. It was the line.

REPO

github.com/PadiusD1/morgan-pantry-store

OPEN TO

Software engineering and finance internships, summer 2026

REACH ME

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FOLLOW FOR THE NEXT BUILD